

CAPE GRIM, Australia

WMO: 959540

Lat: 40.6792S

Lon: 144.6903E

Elev: 285

StdP: 14.54

Time Zone: 10.00 (AUS)

Period: 11-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	42.0	43.6	33.4	28.2	45.4	35.2	30.4	46.3	47.5	48.8	43.0	50.4	14.2	110	0.895

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.8	71.9	65.3	70.0	63.8	68.2	62.9	66.3	69.6	65.1	68.3	64.0	67.1	23.8	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.1	93.9	67.5	63.8	89.6	66.8	62.6	85.8	66.0	31.2	69.6	30.3	68.6	29.4	67.2	73.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
40.7	37.2	33.6	38.4	77.6	1.6	3.7	37.2	80.3	36.2	82.4	35.3	84.5	34.1	87.2
			36.0	69.4	1.5	1.8	34.9	70.6	34.0	71.7	33.2	72.6	32.1	73.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.5	61.2	61.7	60.7	57.6	54.2	51.7	50.9	50.2	51.5	52.9	55.5	58.5
	DBStd	5.23	3.61	3.56	4.01	3.33	3.08	2.93	2.59	2.89	2.86	3.01	3.72	3.63
	HDD50	99	0	0	0	0	2	18	21	35	16	6	1	0
	HDD65	3500	129	105	143	223	335	398	437	459	406	374	286	205
	CDD50	2109	347	326	332	228	133	69	49	40	60	97	165	263
	CDD65	38	12	11	10	1	0	0	0	0	0	0	1	3
	CDH74	31	18	6	5	0	0	0	0	0	0	0	0	2
	CDH80	1	1	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	19.3	19.0	19.8	18.9	17.4	19.1	18.2	21.1	19.7	18.9	19.3	20.2	19.3
Precipitation	PrecAvg	37.2	1.7	1.6	2.2	3.0	3.7	4.1	4.8	4.4	3.7	3.1	2.6	2.1
	PrecMax	51.1	4.4	6.3	6.0	7.1	10.0	7.0	8.8	9.3	8.7	5.9	5.9	5.7
	PrecMin	26.1	0.5	0.2	0.3	1.0	1.1	1.5	1.9	1.1	0.9	1.2	0.8	0.4
	PrecStd	6.0	0.9	1.4	1.4	1.6	2.0	1.3	1.8	1.7	1.6	1.3	1.1	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.4	74.1	74.2	68.1	62.8	58.7	58.3	58.1	60.8	64.5	69.3	72.3
		MCWB	67.0	66.6	66.9	61.2	57.5	55.6	54.9	54.0	55.7	57.4	62.4	64.4
	2%	DB	71.7	71.0	70.7	65.7	61.2	57.5	56.7	56.6	58.9	61.7	66.1	68.9
		MCWB	65.5	64.6	64.6	61.0	57.0	54.7	53.3	53.0	54.7	56.5	61.6	62.4
	5%	DB	69.3	69.3	68.5	63.9	60.1	56.7	55.7	55.6	57.3	59.4	63.3	66.1
		MCWB	63.4	63.5	63.7	60.0	56.5	53.9	52.5	52.2	53.9	55.9	59.8	61.0
	10%	DB	66.9	67.4	66.7	62.7	59.0	56.0	54.8	54.6	55.8	57.7	61.2	64.3
		MCWB	62.1	62.6	62.7	59.5	56.0	53.1	51.7	51.4	53.0	54.7	58.2	60.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.3	67.4	67.7	63.6	59.9	57.3	55.9	54.9	56.7	59.2	64.6	65.4
		MCDB	72.7	72.4	72.5	65.5	61.0	57.8	57.4	56.8	58.7	61.3	66.7	69.8
	2%	WB	65.7	66.0	66.2	62.1	58.2	55.6	54.3	53.8	55.5	57.7	62.1	63.9
		MCDB	69.4	69.0	69.1	63.9	59.8	56.5	55.7	55.8	57.8	60.0	65.1	67.1
	5%	WB	64.4	64.8	64.6	61.1	57.3	54.6	53.3	52.9	54.5	56.4	60.3	62.5
		MCDB	68.5	67.8	67.4	63.1	59.2	56.0	54.7	54.9	56.7	58.8	63.0	65.3
	10%	WB	62.9	63.7	63.1	59.9	56.3	53.6	52.2	52.0	53.4	55.1	58.1	60.7
		MCDB	66.4	66.7	65.9	62.3	58.5	55.5	54.1	54.0	55.4	57.5	61.1	63.6
Mean Daily Temperature Range	5% DB	MDBR	8.4	8.8	8.4	7.3	6.1	5.9	5.7	6.4	6.5	7.5	7.2	8.3
		MCDBR	12.3	10.9	10.9	9.2	6.5	5.7	5.5	6.7	7.7	9.9	10.4	11.6
		MCWBR	7.7	7.4	7.1	6.8	6.0	5.6	5.3	5.9	6.4	7.4	6.8	7.8
	5% WB	MCDBR	10.8	9.2	9.5	7.9	6.5	5.2	5.3	6.2	6.8	9.5	9.7	10.2
		MCWBR	7.5	7.7	6.9	6.4	6.1	5.6	5.7	6.1	6.3	7.6	6.6	7.8
Clear-Sky Solar Irradiance	taub	0.347	0.339	0.332	0.326	0.313	0.306	0.309	0.322	0.359	0.339	0.340	0.340	
	taud	2.528	2.568	2.586	2.578	2.597	2.602	2.598	2.560	2.427	2.512	2.529	2.543	
	Ebn at Noon	312	306	295	274	257	249	256	272	279	303	312	315	
	Edh at Noon	35	32	29	25	21	19	20	25	33	33	34	34	
All-Sky Solar Radiation	RadAvg	2127	1854	1391	950	611	486	531	790	1160	1632	1951	2139	
	RadStd	134	90	66	54	36	36	30	60	65	90	113	140	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (11 neighbors)	+0.52	N/A	+0.76	+1.06	+0.88	+0.92	N/A	-146	+166	+14

Nomenclature: See separate page